2. Case Study:

Now, the number of growing high-rise civil structures is growing throughout the world, and utmost of them use concrete as a structure material. Concrete may lose its strength due to nonstop burden and environmental impacts. As a result, damage (crack and flake) may do on the surface face of the structure. Whenever these flaws are left unexplored and undressed, the integrity of the structure may be compromised. Thus, regular conservation of the structure is demanded. Some former vision- grounded studies have used a drone as a vehicle to capture and record the current state of the structure. Latterly, captured videos and images were anatomized to determine damage using object bracket, localization, and segmentation styles. Also, in a many studies, drones bear the collected data to the server using a wireless medium. Still, the developed systems are veritably complex, time-intensive, and bear high bandwidth. This paper proposes a crack discovery system that's based on UAVs and the VGG16 algorithm to quantify the length and range of cracks. The rapid-fire development of UAV technology has greatly bettered the effectiveness of the discovery of concrete ground cracks. It's anticipated to introduce robots/ drones similar as Unmanned Aerial Vehicles (UAV) for the inspection because it can reduce the burden on architects and collect data from the spaces where people are delicate to approach. Still, it's frequently delicate to secure sufficient accurateness for the data collected by robots. This problem can be answered by conforming the shooting angle and distance if possible to confirm the discovery delicacy of the crack contained in the photographed image on the spot. Also, drones examination is available as a useful tool that can realize the real- time discovery. In our design of crack discovery in concrete through AI drones we use image processing to prepare the images for analysis by applying preprocessing ways, similar as resizing, normalization, and color adaptations. The training dataset collection was used to train the model and the confirmation dataset was used during training to cover the model learning wind. The datasets contains images of colorful concrete shells with and without crack. The image data are divided into two as negative (without crack) and positive (with crack) in separate brochure for image classification. Control RGB, these images featured values for each of the primary colours on the spectrum red, green, and blue, also known as 3-channels (control RGB). A combination of these three can produce all possible colour pallets. The most common system of DL for crack discovery is CNN. CNNs have been shown to out-perform edge discovery and ML classifiers. Pre-trained CNNs have preliminarily demonstrated a delicacy of over 98 for crack/non-crack discovery. MATLAB used to remove the noise from the atmosphere and python used in coding, The Keras package contains numerous DL infrastructures. The Keras VGG16 model is grounded on the model proposed. A non-crack was labelled 0, while a crack was labelled 1.

3. Introduction:

The AI crack detection system stands as a technological breakthrough poised to revolutionize maintenance and security. This pioneering initiative addresses the growing need for a more sophisticated, efficient, and proactive approach to identifying, assessing and managing cracks in different types of buildings such as the increasing reliance of society on complex infrastructure ensuring their integrity becomes paramount. But traditional methods often fail to detect and accurately assess cracks, leading to potential hazards and ineffective maintenance.

Scope:

The AI crack detection system covers a wide range of applications from bridges and roads to buildings and other critical structures and aims to revolutionize maintenance practices through artificial intelligence and recognition technology the wonderful to be used. This system goes beyond mere crack detection, providing local accuracy, severity assessment and real-time monitoring, all while ensuring compliance with data security and regulatory standards.

Problem Statement:

Current maintenance systems are often based on time-consuming manual inspections, prone to human error, insufficient to detect subtle but important cracks This increases safety risks, architectural vulnerabilities, and maintenance costs. The challenge is to develop comprehensive, automated solutions that can accurately and quickly detect, measure, and monitor cracks in various buildings and environments.

Program Goals and Objectives:

The main objective of an AI crack detection system is to promote a quick and accurate repair method. Its objectives are:

1. **Accurate detection and positioning:** Use advanced algorithms and sensor technology to accurately detect and navigate cracks in buildings.
2. **Assessment and severity classification**: To develop an algorithm for accurate severity assessment and classification of detected cracks to prioritize repair efforts.
3. **Real-time analytics and reporting:** Enable real-time analytics using agile reporting channels to ensure prompt action in the event of critical events.

4. **Data Security and Compliance:** Implement strong security measures to protect sensitive data and ensure compliance with legal and regulatory standards.
5. **Data Management and Continuous Improvement**: Simplify data management processes to ensure consistent system accuracy and efficiency.

Motivation:

The motivation behind the AI Crack Detection System comes from a shared goal of improving safety and efficiency in the industry. By integrating advanced technologies, the system aims to reduce risks, reduce maintenance costs, and improve overall system integrity. The ability to save lives, resources, and time by ensuring the longevity of critical structures is a driving force in the development and implementation of this underground system.

4. Scope Problem:

Planning Document:

In this project, we are going to make a crack detection in concrete structure through AI for the purpose of Infrastructure, such as buildings, bridges, pavement, etc., needs to be examined periodically to maintain its reliability and structural health. Visual signs of cracks and depressions indicate stress and wear and tear over time, leading to failure/collapse if these cracks are located at critical locations, such as in load-bearing joints. Manual inspection is carried out by experienced inspectors who require long inspection times and rely on their empirical and subjective knowledge. We will work in this project by using Software Development Life Cycle. After analyzing the objectives, the plan for this project is as follows:

Scope Planning:

Following the objectives of the project, detecting surface cracks in concrete this service will be provided through a web based app. All the team members got together, we define the project's objectives, requirements, and deliverables. Some of the features and functionalities the software should include like **crack detection**, **convolutional neural network**, **image processing**, **deep learning**, **damage detection**, **machine learning** that we will build in this project.

Problem Solved by the Project:

Manual examination is carried out by educated inspectors who bear long inspection times and calculate on their empirical and private knowledge. This lengthy process results in waits that further compromise the infrastructure’s structural integrity. To address this limitation, this study proposes a deep learning (DL)-grounded independent crack discovery system using the convolutional neural network (CNN) system. To better the CNN classification performance for enhanced pixel segmentation, 40,000 RGB images were reprocessed before training a pre-trained VGG16 architecture to produce different CNN models.

Gather Requirements:

Amna and Maham both work together and gather the detailed requirements from the stakeholder. The requirement should be specific, measurable, achievable, relevant, and time-bound.

Resources Management:

We have planned and estimate the efforts, time, and resources (including human resources, machine resources, and software). Our team will determine and plan the capabilities as well as the availability of our resources. We have planned how we can compare the required skills with the available expertise of these resources.

Team:

We have planned according to available team resources. The project is divided into small phases and module. We both will perform our part of work.

Quality Assurance Processes:

For quality assurance processes, we include code reviews, testing plans, and validation procedures, to ensure the software meets the required quality standards.

Cost Estimating:

The cost for the project was determined by both the development team and the stakeholder. All the functionalities and features that the stakeholder wants in their project were added according to the budget. The development team makes adjustments and manages all phases to ensure that any risks that arise during development are managed within the budget.

Review and Adjust:

We continuously review project progress, compare it to the original plan, and adjust as necessary. We are prepared to adapt to changing circumstances and unexpected challenges.

Timetable and schedules:

The timetable has been planned. The team had started the work for planning of 1st and 2nd phases in the month of April. 3rd phase was started in May. 4th and 5th phases was completed in the end of May. The last phase was completed before mid of the month of June.

5. Elicitation Techniques:

1. Brainstorming
2. Mind-mapping
3. Interviews

6. Requirement Elicitation Technique Details and Results:

Brainstorming:

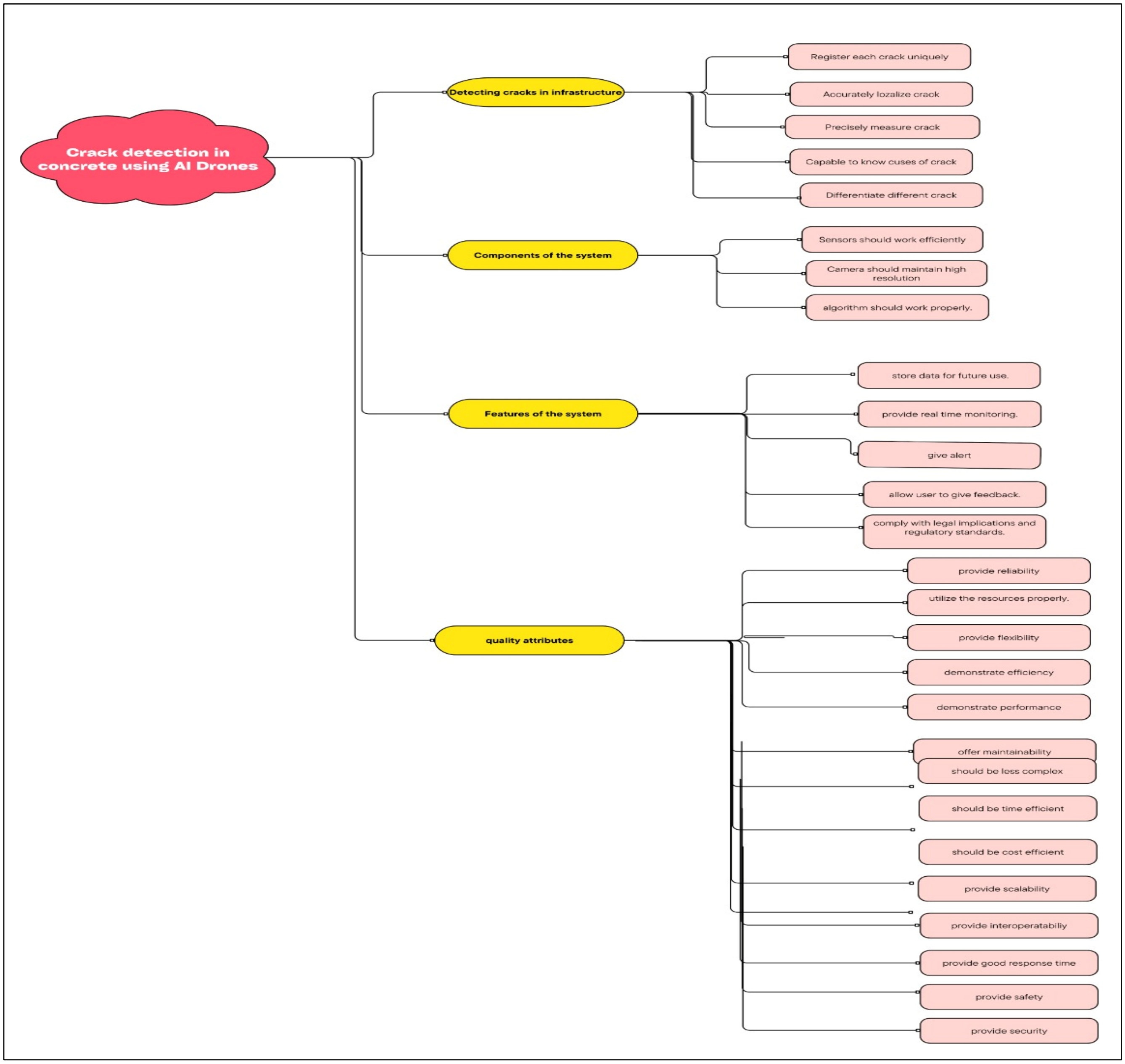
The brainstorming process in developing an AI crack detection system is an important step in the requirements gathering and development phase. Brainstorming is a collaborative process for discussing, analysing, and refining ideas, features and strategies which is important in the design and operation of the system. We conducted a brainstorming session among our group members. The following are the main points from the brainstorming session:

1. According to Amna some people prefer manual crack detection system.
2. Maham further added that Crack detection system using AI drones is a better option.
3. Then they discussed the functional requirements of the system in which Maham said this system be helpful to detect the cracks in infrastructure (like buildings, bridges, dams).
4. To which Amna added then the system should accurately localize the detected cracks.
5. Maham added that the system should precisely measure the detected cracks and give details about the size, length, and depth of the identified crack.
6. Amna added the system shall be capable to know causes of cracks.
7. Maham suggested that the camera should effectively capture a wide distance range.
8. Amna added that the system should maintain high-resolution imagery for crack detection purposes.

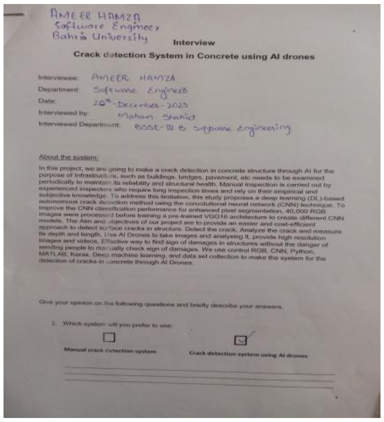
9. Maham suggested that the system account for changes in temperature, humidity, or other environmental factors.
10. Amna added that the system shall be able to differentiate between different type of cracks.
11. Maham proposed that the system shall provide the feature of alert and reporting the team.
12. Amna suggested the system shall provide Sensors for proper crack detection.
13. Amna recommended that the system shall provide real time monitoring.
14. Maham added the system shall store data for future use.
15. Amna suggested that the system shall ensure each crack is registered and processed uniquely.
16. Maham added that the system should allow users to give feedback.
17. Then they started their discussion about the non- functional requirements of the system in which Amna said the system shall maintain operational reliability in various weather conditions (like low breeze, cloudy, sunny, drizzle).
18. To which Maham added the system should demonstrate efficiency in its operations.
19. Amna suggested that the system should be flexible to changes.
20. Maham recommended the system should maintain high performance.
21. To which Amna added that the system shall offer maintainability.
22. Amna suggested that the system should be cost efficient.
23. Maham added that the system shall be time efficient.
24. Amna proposed that the system should be less complex.
25. Maham recommended that the system shall provide scalability features.
26. Amna proposed the system should utilize the resources properly.
27. Maham added that response time of the system should be good.
28. Amna suggested that the system should be interoperate-able.
29. Maham recommended system should ensure safety of the inspection team.
30. Amna added that the system shall comply with legal implications and regulatory standards.
31. Amna proposed that the system should provide security.

32. Maham recommended that the system should be user-friendly.

Mind-Mapping:



Interviews:



AMEER HAMZA
Software Engineer
Bahria University

Interview

Crack detection System in Concrete using AI drones

Interviewee: AMEER HAMZA
Department: Software Engineer
Date: 20th-December-2023
Interviewed by: Maham Shahid
Interviewed Department: BSSE-III B Software Engineering

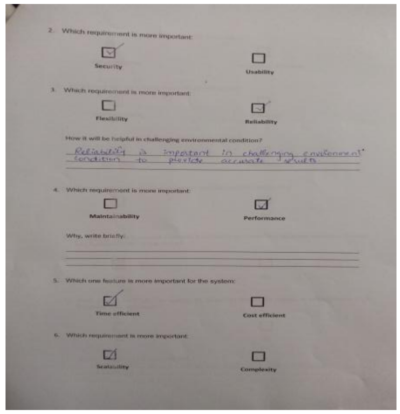
About the system:

In this project, we are going to make a crack detection in concrete structure through AI for the purpose of Infrastructure, such as buildings, bridges, pavement, etc needs to be examined periodically to maintain its reliability and structural health. Manual inspection is carried out by experienced inspectors who require long inspection times and rely on their empirical and subjective knowledge. To address this limitation, this study proposes a deep learning (DL)-based autonomous crack detection method using the convolutional neural network (CNN) technique. To improve the CNN classification performance for enhanced pixel segmentation, 40,000 RGB images were processed before training a pre-trained VGG16 architecture to create different CNN models. The Aim and objectives of our project are to provide an easier and cost-efficient approach to detect surface cracks in structure. Detect the crack, Analyze the crack and measure its depth and length, Use AI Drones to take images and analysing it, provide high resolution images and videos, Effective way to find sign of damages in structures without the danger of sending people to manually check sign of damages. We use control RGB, CNN, Python, MATLAB, Keras, Deep machine learning, and data set collection to make the system for the detection of cracks in concrete through AI Drones.

Give your opinion on the following questions and briefly describe your answers.

1. Which system will you prefer to use:

Manual crack detection system | Crack detection system using AI drones



2. Which requirement is more important:

Security | Usability

3. Which requirement is more important:

Flexibility | Reliability

How it will be helpful in challenging environmental condition?

Reliability is important in challenging environment condition to provide accurate results.

4. Which requirement is more important:

Maintainability | Performance

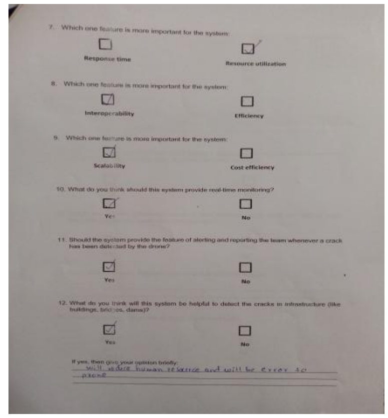
Why, write briefly:

5. Which one feature is more important for the system:

Time efficient | Cost efficient

6. Which requirement is more important:

Scalability | Complexity



7. Which one feature is more important for the system:

Response time | Resource utilization

8. Which one feature is more important for the system:

Interoperability | Efficiency

9. Which one feature is more important for the system:

Scalability | Cost efficiency

10. What do you think should this system provide real-time monitoring?

Yes | No

11. Should the system provide the feature of alerting and reporting the team whenever a crack has been detected by the drone?

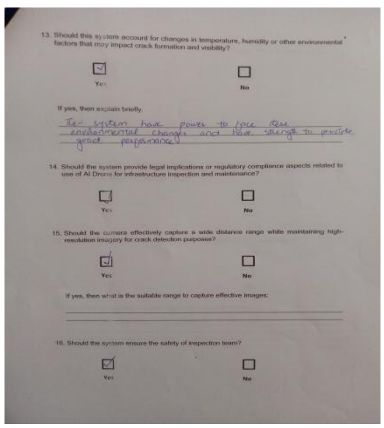
Yes | No

12. What do you think will this system be helpful to detect the cracks in infrastructure (like buildings, bridges, dams)?

Yes | No

If yes, then give your opinion briefly:

will reduce human resource and will be error to prone



13. Should this system account for changes in temperature, humidity or other environmental factors that may impact crack formation and visibility?

Yes | No

If yes, then explain briefly.

The system have power to face these environmental changes and have strength to provide good performance.

14. Should the system provide legal implications or regulatory compliance aspects related to use of AI Drone for infrastructure inspection and maintenance?

Yes | No

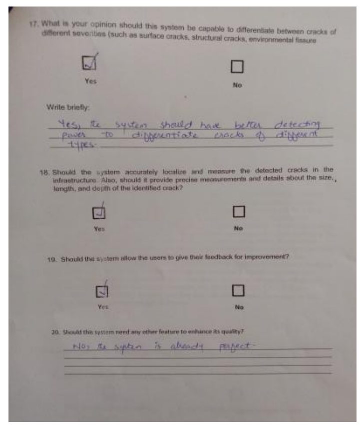
15. Should the camera effectively capture a wide distance range while maintaining high-resolution imagery for crack detection purposes?

Yes | No

If yes, then what is the suitable range to capture effective images:

16. Should the system ensure the safety of inspection team?

Yes | No



17. What is your opinion should this system be capable to differentiate between cracks of different severities (such as surface cracks, structural cracks, environmental fissure

Yes | No

Write briefly:

Yes, the system should have better detecting power to differentiate cracks of different types.

18. Should the system accurately localize and measure the detected cracks in the infrastructure. Also, should it provide precise measurements and details about the size, length, and depth of the identified crack?

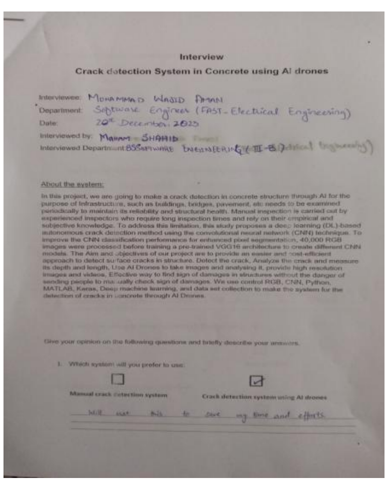
Yes | No

19. Should the system allow the users to give their feedback for improvement?

Yes | No

20. Should this system need any other feature to enhance its quality?

No, the system is already perfect.



Interview

Crack detection System in Concrete using AI drones

Interviewee: MOHAMMAD WAJID AMAN
Department: Software Engineer (FAST-Electrical Engineering)
Date: 20th December 2023
Interviewed by: MAHAM SHAHID
Interviewed Department: BSSOFTWARE ENGINEERING (III-B)

About the system:

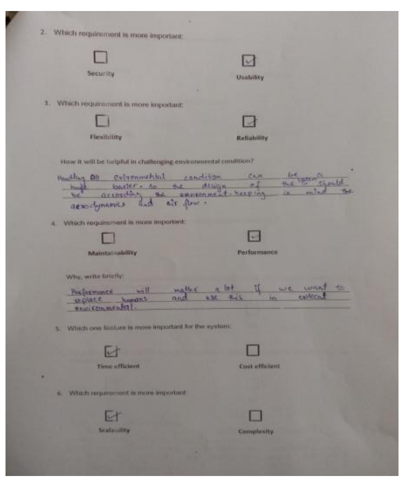
In this project, we are going to make a crack detection in concrete structure through AI for the purpose of Infrastructure, such as buildings, bridges, pavement, etc needs to be examined periodically to maintain its reliability and structural health. Manual inspection is carried out by experienced inspectors who require long inspection times and rely on their empirical and subjective knowledge. To address this limitation, this study proposes a deep learning (DL)-based autonomous crack detection method using the convolutional neural network (CNN) technique. To improve the CNN classification performance for enhanced pixel segmentation, 40,000 RGB images were processed before training a pre-trained VGG16 architecture to create different CNN models. The Aim and objectives of our project are to provide an easier and cost-efficient approach to detect surface cracks in structure. Detect the crack, Analyze the crack and measure its depth and length, Use AI Drones to take images and analysing it, provide high resolution images and videos, Effective way to find sign of damages in structures without the danger of sending people to manually check sign of damages. We use control RGB, CNN, Python, MATLAB, Keras, Deep machine learning, and data set collection to make the system for the detection of cracks in concrete through AI Drones.

Give your opinion on the following questions and briefly describe your answers.

1. Which system will you prefer to use:

Manual crack detection system | Crack detection system using AI drones

Will use this to save my time and efforts.



2. Which requirement is more important:

Security | Usability

3. Which requirement is more important:

Flexibility | Reliability

How it will be helpful in challenging environmental condition?

Handling all environmental condition can be a huge barrier. So the design of the system should be according the environment keeping in mind the aerodynamics and air flow.

4. Which requirement is more important:

Maintainability | Performance

Why, write briefly:

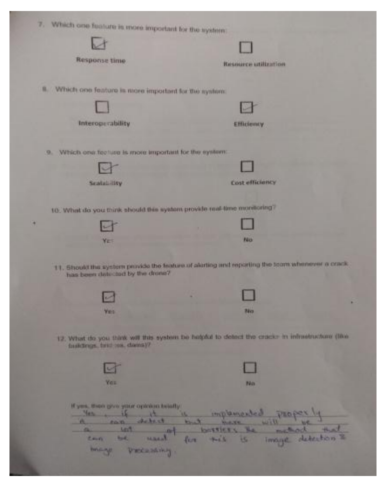
Performance will matter a lot if we want to replace humans and use this in critical environmental.

5. Which one feature is more important for the system:

Time efficient | Cost efficient

6. Which requirement is more important:

Scalability | Complexity



7. Which one feature is more important for the system:

Response time | Resource utilization

8. Which one feature is more important for the system:

Interoperability | Efficiency

9. Which one feature is more important for the system:

Scalability | Cost efficiency

10. What do you think should this system provide real-time monitoring?

Yes | No

11. Should the system provide the feature of alerting and reporting the team whenever a crack has been detected by the drone?

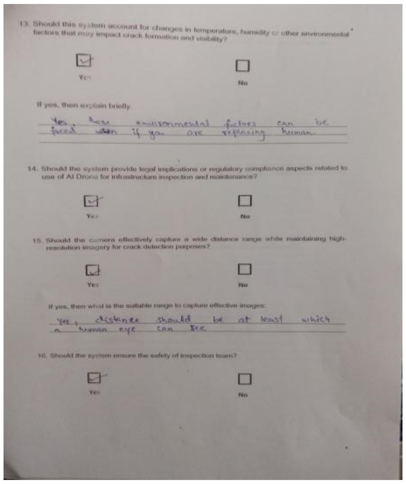
Yes | No

12. What do you think will this system be helpful to detect the cracks in infrastructure (like buildings, bridges, dams)?

Yes | No

If yes, then give your opinion briefly:

Yes, if it is implemented properly it can detect but there will be a lot of barriers. The method that can be used for this is image detection & image processing.



13. Should this system account for changes in temperature, humidity or other environmental factors that may impact crack formation and visibility?

☑ Yes ☐ No

If yes, then explain briefly.

Yes, [illegible] environmental factors can be faced [illegible] if you are replacing human.

14. Should the system provide legal implications or regulatory compliance aspects related to use of AI Drone for infrastructure inspection and maintenance?

☑ Yes ☐ No

15. Should the camera effectively capture a wide distance range while maintaining high-resolution imagery for crack detection purposes?

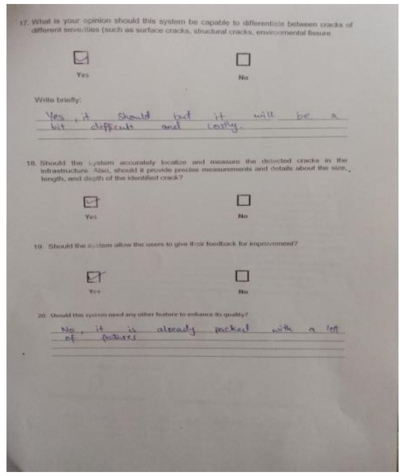
☑ Yes ☐ No

If yes, then what is the suitable range to capture effective images:

Yes, distance should be at least which a human eye can see

16. Should the system ensure the safety of inspection team?

☑ Yes ☐ No



17. What is your opinion should this system be capable to differentiate between cracks of different severities (such as surface cracks, structural cracks, environmental fissure

☑ Yes ☐ No

Write briefly:

Yes, it should but it will be a bit difficult and costly.

18. Should the system accurately localize and measure the detected cracks in the infrastructure. Also, should it provide precise measurements and details about the size, length, and depth of the identified crack?

☑ Yes ☐ No

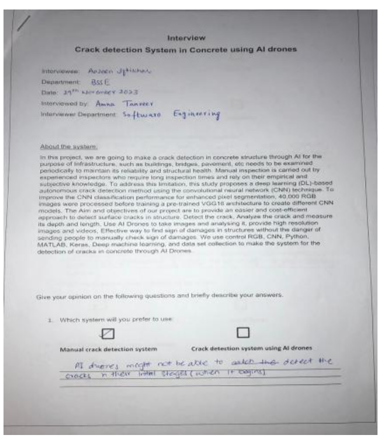
19. Should the system allow the users to give their feedback for improvement?

☑ Yes ☐ No

20. Should this system need any other feature to enhance its quality?

No, it is already packed with a lot of features

Conducted from Senior Students of Software Engineering:



Interview

Crack detection System in Concrete using AI drones

Interviewee: Aqsa Iftikhar

Department: BSSE

Date: 29th November 2023

Interviewed by: Amna Tanveer

Interviewer Department: Software Engineering

About the system:

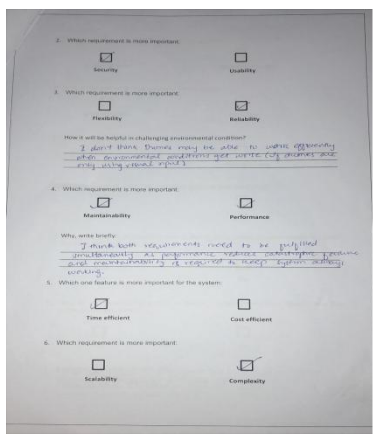
In this project, we are going to make a crack detection in concrete structure through AI for the purpose of Infrastructure, such as buildings, bridges, pavement, etc needs to be examined periodically to maintain its reliability and structural health. Manual inspection is carried out by experienced inspectors who require long inspection times and rely on their empirical and subjective knowledge. To address this limitation, this study proposes a deep learning (DL)-based autonomous crack detection method using the convolutional neural network (CNN) technique. To improve the CNN classification performance for enhanced pixel segmentation, 40,000 RGB images were processed before training a pre-trained VGG16 architecture to create different CNN models. The Aim and objectives of our project are to provide an easier and cost-efficient approach to detect surface cracks in structure. Detect the crack, Analyze the crack and measure its depth and length. Use AI Drones to take images and analysing it, provide high resolution images and videos, Effective way to find sign of damages in structures without the danger of sending people to manually check sign of damages. We use control RGB, CNN, Python, MATLAB, Keras, Deep machine learning, and data set collection to make the system for the detection of cracks in concrete through AI Drones.

Give your opinion on the following questions and briefly describe your answers.

1. Which system will you prefer to use:

☑ Manual crack detection system ☐ Crack detection system using AI drones

AI drones might not be able to ~~catch the~~ detect the cracks in their initial stages (when it begins)



2. Which requirement is more important:

☑ Security ☐ Usability

3. Which requirement is more important:

☐ Flexibility ☑ Reliability

How it will be helpful in challenging environmental condition?

I don't think Drones may be able to work efficiently when environmental conditions get worse (if drones are only using visual input)

4. Which requirement is more important:

☑ Maintainability ☑ Performance

Why, write briefly:

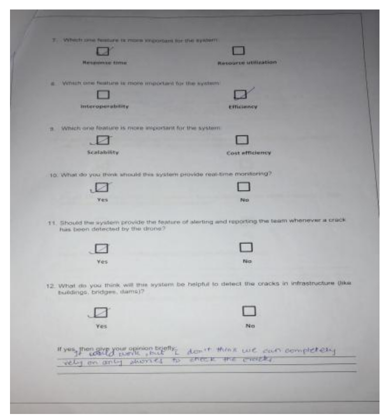
I think both requirements need to be fulfilled simultaneously as performance reduces catastrophic [illegible] and maintainability is required to keep system actively working.

5. Which one feature is more important for the system:

☑ Time efficient ☐ Cost efficient

6. Which requirement is more important:

☐ Scalability ☑ Complexity



7. Which one feature is more important for the system:

☑ Response time ☐ Resource utilization

8. Which one feature is more important for the system:

☐ Interoperability ☑ Efficiency

9. Which one feature is more important for the system:

☑ Scalability ☐ Cost efficiency

10. What do you think should this system provide real-time monitoring?

☑ Yes ☐ No

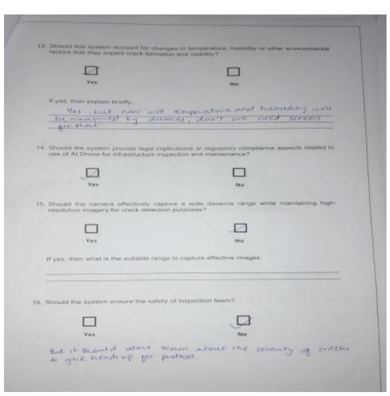
11. Should the system provide the feature of alerting and reporting the team whenever a crack has been detected by the drone?

☑ Yes ☐ No

12. What do you think will this system be helpful to detect the cracks in infrastructure (like buildings, bridges, dams)?

☑ Yes ☐ No

If yes, then give your opinion briefly: It would work, but I don't think we can completely rely on only drones to check the cracks.



13. Should this system account for changes in temperature, humidity or other environmental factors that may impact crack formation and visibility?

☑ Yes ☐ No

If yes, then explain briefly.

Yes, but how will temperature and humidity will be measured by drones, don't we need sensors for that.

14. Should the system provide legal implications or regulatory compliance aspects related to use of AI Drone for infrastructure inspection and maintenance?

☑ Yes ☐ No

15. Should the camera effectively capture a wide distance range while maintaining high-resolution imagery for crack detection purposes?

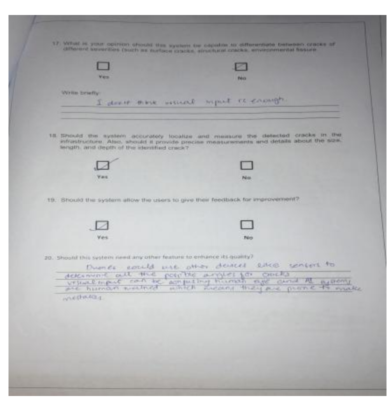
☐ Yes ☑ No

If yes, then what is the suitable range to capture effective images:

16. Should the system ensure the safety of inspection team?

☐ Yes ☑ No

But it should alert team about the severity of cracks to give heads up for protocol.



17. What is your opinion should this system be capable to differentiate between cracks of different severities (such as surface cracks, structural cracks, environmental fissure

☐ Yes ☑ No

Write briefly:

I don't think visual input is enough.

18. Should the system accurately localize and measure the detected cracks in the infrastructure. Also, should it provide precise measurements and details about the size, length, and depth of the identified crack?

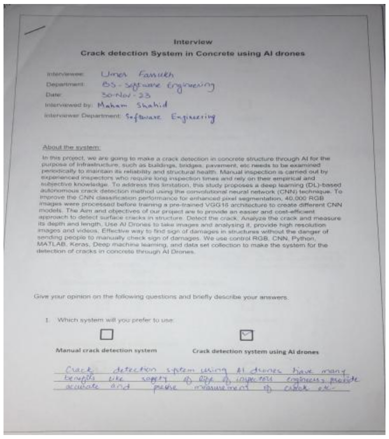
☑ Yes ☐ No

19. Should the system allow the users to give their feedback for improvement?

☑ Yes ☐ No

20. Should this system need any other feature to enhance its quality?

Drones could use other devices like sensors to determine all the possible angles for cracks. Visual input can be [illegible] human eye and AI systems are human trained which means they are prone to make mistakes.



Interview

Crack detection System in Concrete using AI drones

Interviewee: Umer Farrukh
Department: BS - Software Engineering
Date: 30-Nov-23
Interviewed by: Maham Shahid
Interviewer Department: Software Engineering

About the system:

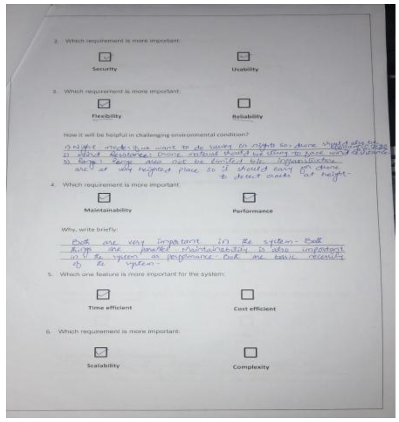
In this project, we are going to make a crack detection in concrete structure through AI for the purpose of infrastructure, such as buildings, bridges, pavement, etc needs to be examined periodically to maintain its reliability and structural health. Manual inspection is carried out by experienced inspectors who require long inspection times and rely on their empirical and subjective knowledge. To address this limitation, this study proposes a deep learning (DL)-based autonomous crack detection method using the convolutional neural network (CNN) technique. To improve the CNN classification performance for enhanced pixel segmentation, 40,000 RGB images were processed before training a pre-trained VGG16 architecture to create different CNN models. The Aim and objectives of our project are to provide an easier and cost-efficient approach to detect surface cracks in structure. Detect the crack. Analyze the crack and measure its depth and length. Use AI Drones to take images and analysing it, provide high resolution images and videos. Effective way to find sign of damages in structures without the danger of sending people to manually check sign of damages. We use control RGB, CNN, Python, MATLAB, Keras, Deep machine learning, and data set collection to make the system for the detection of cracks in concrete through AI Drones.

Give your opinion on the following questions and briefly describe your answers.

1. Which system will you prefer to use:

☐ Manual crack detection system ☑ Crack detection system using AI drones

Crack detection system using AI drones have many benefits like safety of life of inspectors engineers, provide accurate and precise measurement of crack etc.



2. Which requirement is more important:

☑ Security ☑ Usability

3. Which requirement is more important:

☑ Flexibility ☐ Reliability

How it will be helpful in challenging environmental condition?

1) Night mode: If we want to do survey in night so, drone should also [illegible]
2) Wind Resistance: Drone material should be strong to face wind resistance
3) Range: Range also not be limited b/c infrastructure are at very heighted place so it should easy for drone to detect cracks at height.

4. Which requirement is more important:

☑ Maintainability ☑ Performance

Why, write briefly:

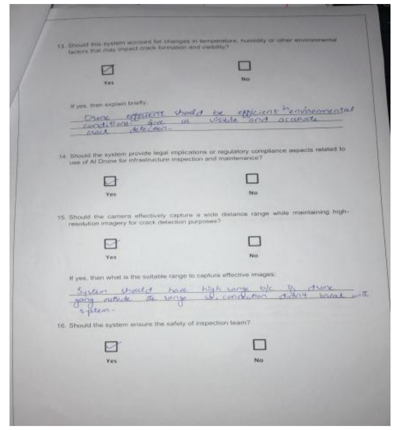
Both are very important in the system. Both things are parallel. Maintainability is also important in the system as performance. Both are basic necessity of the system.

5. Which one feature is more important for the system:

☑ Time efficient ☐ Cost efficient

6. Which requirement is more important:

☑ Scalability ☐ Complexity



13. Should this system account for changes in temperature, humidity or other environmental factors that may impact crack formation and visibility?

☑ Yes ☐ No

If yes, then explain briefly:

Drone efficient should be efficient in environmental conditions. Give us visible and accurate crack detection.

14. Should the system provide legal implications or regulatory compliance aspects related to use of AI Drone for infrastructure inspection and maintenance?

☑ Yes ☐ No

15. Should the camera effectively capture a wide distance range while maintaining high-resolution imagery for crack detection purposes?

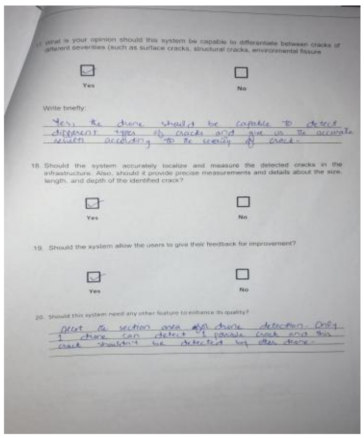
☑ Yes ☐ No

If yes, then what is the suitable range to capture effective images:

System should have high range b/c if drone going outside the range so, connection didn't break with system.

16. Should the system ensure the safety of inspection team?

☑ Yes ☐ No



17. What is your opinion should this system be capable to differentiate between cracks of different severities (such as surface cracks, structural cracks, environmental fissure

☑ Yes ☐ No

Write briefly:

Yes, the drone should be capable to detect different types of cracks and give us the accurate results according to the severity of crack.

18. Should the system accurately localize and measure the detected cracks in the infrastructure. Also, should it provide precise measurements and details about the size, length, and depth of the identified crack?

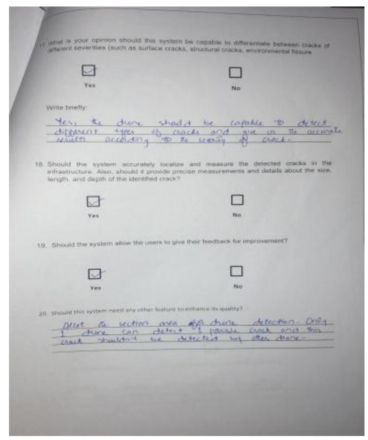
☑ Yes ☐ No

19. Should the system allow the users to give their feedback for improvement?

☑ Yes ☐ No

20. Should this system need any other feature to enhance its quality?

Detect the section area [illegible] drone detection. Only 1 drone can detect 1 possible crack and this crack shouldn't be detected by other drone.



17. What is your opinion should this system be capable to differentiate between cracks of different severities (such as surface cracks, structural cracks, environmental fissure

☑ Yes ☐ No

Write briefly:

Yes, the drone should be capable to detect different types of cracks and give us the accurate results according to the severity of crack.

18. Should the system accurately localize and measure the detected cracks in the infrastructure. Also, should it provide precise measurements and details about the size, length, and depth of the identified crack?

☑ Yes ☐ No

19. Should the system allow the users to give their feedback for improvement?

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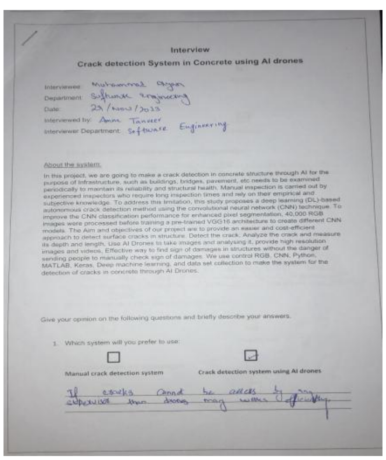
Detect the section area [illegible] drone detection. Only 1 drone can detect 1 possible crack and this crack shouldn't be detected by other drone.



Interview

Crack detection System in Concrete using AI drones

Interviewee: Muhammad [illegible]
Department: Software Engineering
Date: 23/Nov/2023
Interviewed by: Amna Tanveer
Interviewer Department: Software Engineering

About the system:

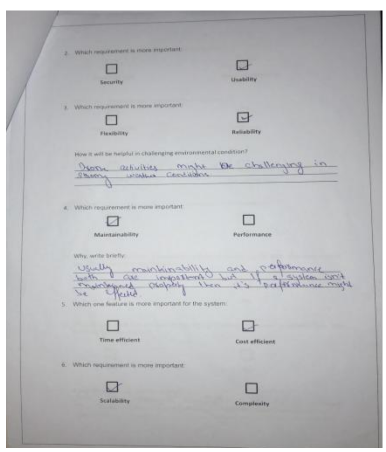
In this project, we are going to make a crack detection in concrete structure through AI for the purpose of infrastructure, such as buildings, bridges, pavement, etc needs to be examined periodically to maintain its reliability and structural health. Manual inspection is carried out by experienced inspectors who require long inspection times and rely on their empirical and subjective knowledge. To address this limitation, this study proposes a deep learning (DL)-based autonomous crack detection method using the convolutional neural network (CNN) technique. To improve the CNN classification performance for enhanced pixel segmentation, 40,000 RGB images were processed before training a pre-trained VGG16 architecture to create different CNN models. The Aim and objectives of our project are to provide an easier and cost-efficient approach to detect surface cracks in structure. Detect the crack. Analyze the crack and measure its depth and length. Use AI Drones to take images and analysing it, provide high resolution images and videos. Effective way to find sign of damages in structures without the danger of sending people to manually check sign of damages. We use control RGB, CNN, Python, MATLAB, Keras, Deep machine learning, and data set collection to make the system for the detection of cracks in concrete through AI Drones.

Give your opinion on the following questions and briefly describe your answers.

1. Which system will you prefer to use:

☐ Manual crack detection system ☑ Crack detection system using AI drones

If cracks cannot be access by any [illegible] then drones may works efficiently.



2. Which requirement is more important:

☐ Security ☑ Usability

3. Which requirement is more important:

☐ Flexibility ☑ Reliability

How it will be helpful in challenging environmental condition?

Drone activities might be challenging in stormy weather conditions

4. Which requirement is more important:

☑ Maintainability ☐ Performance

Why, write briefly:

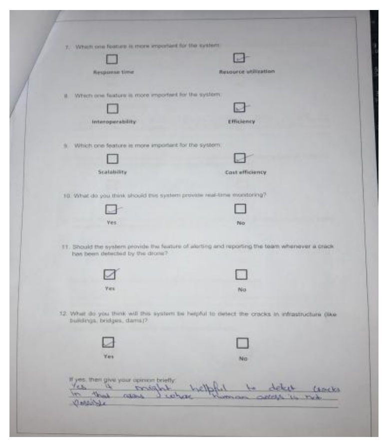
Usually maintainability and performance both are important but if a system isn't maintained properly then it's performance might be effected.

5. Which one feature is more important for the system:

☐ Time efficient ☑ Cost efficient

6. Which requirement is more important:

☑ Scalability ☐ Complexity



7. Which one feature is more important for the system:

☐ Response time ☑ Resource utilization

8. Which one feature is more important for the system:

☐ Interoperability ☑ Efficiency

9. Which one feature is more important for the system:

☐ Scalability ☑ Cost efficiency

10. What do you think should this system provide real-time monitoring?

☑ Yes ☐ No

11. Should the system provide the feature of alerting and reporting the team whenever a crack has beem detected by the drone?

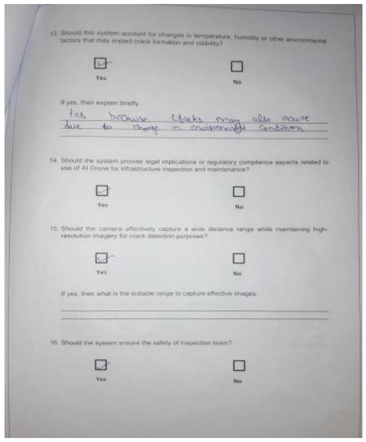
☑ Yes ☐ No

12. What do you think will this system be helpful to detect the cracks in infrastructure (like buildings, bridges, dams)?

☑ Yes ☐ No

If yes, then give your opinion briefly:

Yes it might helpful to detect cracks in that areas where human access is not possible


13. Should this system account for changes in temperature, humidity or other environmental factors that may impact crack formation and visibility?

☑ Yes ☐ No

If yes, then explain briefly.

Yes because cracks may also occure due to change in environmental conditions

14. Should the system provide legal implications or regulatory compliance aspects related to use of AI Drone for infrastructure inspection and maintenance?

☑ Yes ☐ No

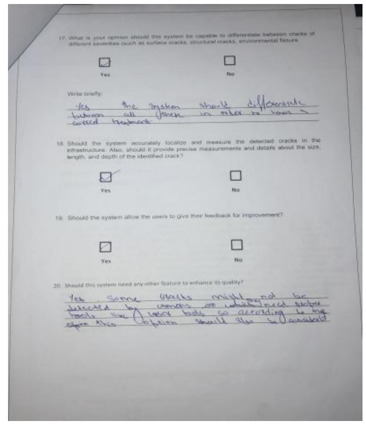
15. Should the camera effectively capture a wide distance range while maintaining high-resolution imagery for crack detection purposes?

☑ Yes ☐ No

If yes, then what is the suitable range to capture effective images.

16. Should the system ensure the safety of inspection team?

☑ Yes ☐ No


17. What is your opinion should this system be capable to differentiate between cracks of different severities (such as surface cracks, structural cracks, environmental fissure

☑ Yes ☐ No

Write briefly:

Yes the system should differentiate between all these in order to have a correct treatment

18. Should the system accurately localize and measure the detected cracks in the infrastructure. Also, should it provide precise measurements and details about the size, length, and depth of the identified crack?

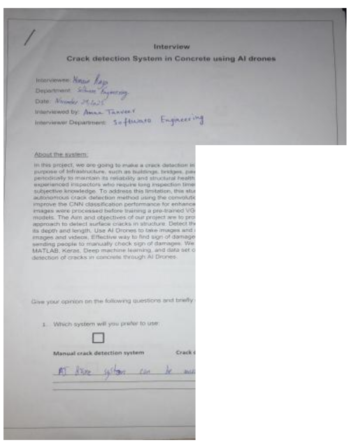
☑ Yes ☐ No

19. Should the system allow the users to give their feedback for improvement?

☑ Yes ☐ No

20. Should this system need any other feature to enhance its quality?

Yes some cracks might not be detected by cameras or [illegible] need [illegible] tools like laser tools so according to me this option should also be considered


Interview

Crack detection System in Concrete using AI drones

Interviewee: [illegible]

Department: Software Engineering

Date: November 29, 2025

Interviewed by: Amna Tanveer

Interviewer Department: Software Engineering

About the system:

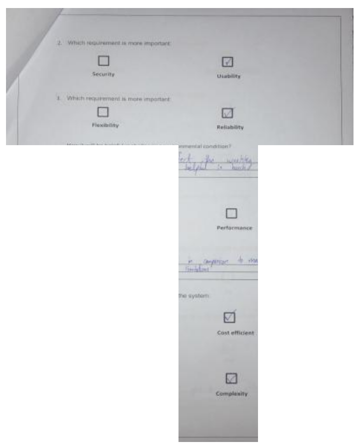
In this project, we are going to make a crack detection in … purpose of Infrastructure, such as buildings, bridges, pa… periodically to maintain its reliability and structural health… experienced inspectors who require long inspection time… subjective knowledge. To address this limitation, this stu… autonomous crack detection method using the convoluti… improve the CNN classification performance for enhance… images were processed before training a pre-trained VG… models. The Aim and objectives of our project are to pro… approach to detect surface cracks in structure. Detect th… its depth and length. Use AI Drones to take images and … images and videos. Effective way to find sign of damage… sending people to manually check sign of damages. We… MATLAB, Keras, Deep machine learning, and data set o… detection of cracks in concrete through AI Drones.

Give your opinion on the following questions and briefly…

1. Which system will you prefer to use:

☐ Manual crack detection system Crack…

AI Drone system can be …


2. Which requirement is more important:

☐ Security ☑ Usability

3. Which requirement is more important:

☐ Flexibility ☑ Reliability

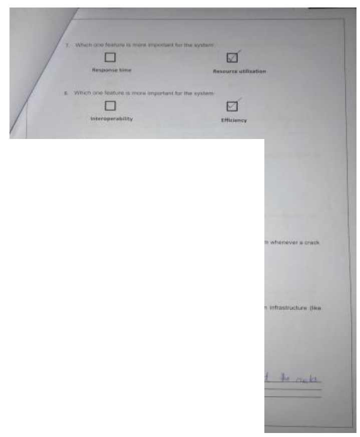
…mental condition?

☐ Performance

…the system:

☑ Cost efficient

☑ Complexity


7. Which one feature is more important for the system:

☐ Response time ☑ Resource utilisation

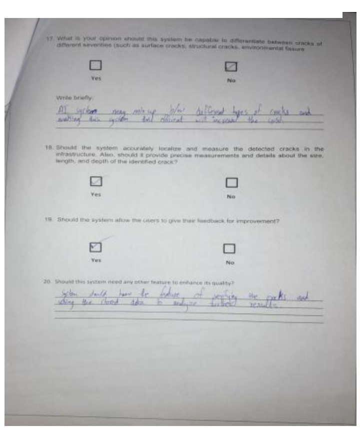
8. Which one feature is more important for the system:

☐ Interoperability ☑ Efficiency

…n whenever a crack

…n infrastructure (like

… the cracks


17. What is your opinion should this system be capable to differentiate between cracks of different severities (such as surface cracks, structural cracks, environmental fissure

☐ Yes ☑ No

Write briefly:

AI system may mix up b/w different types of cracks and making this system that efficient will increase the cost.

18. Should the system accurately localize and measure the detected cracks in the infrastructure. Also, should it provide precise measurements and details about the size, length, and depth of the identified crack?

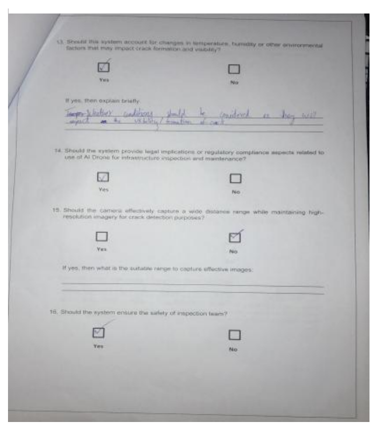
☑ Yes ☐ No

19. Should the system allow the users to give their feedback for improvement?

☑ Yes ☐ No

20. Should this system need any other feature to enhance its quality?

System should have the feature of verifying the cracks and using the stored data to analyze further results.


13. Should this system account for changes in temperature, humidity or other environmental factors that may impact crack formation and visibility?

☑ Yes ☐ No

If yes, then explain briefly.

Weather conditions should be considered as they will impact the visibility / formation of crack

14. Should the system provide legal implications or regulatory compliance aspects related to use of AI Drone for infrastructure inspection and maintenance?

☑ Yes ☐ No

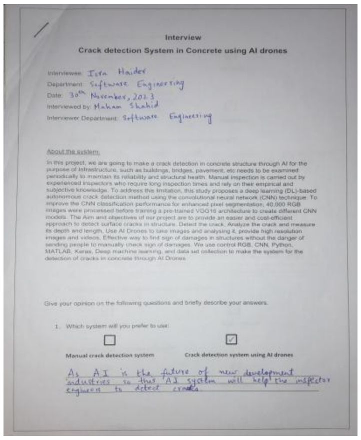
15. Should the camera effectively capture a wide distance range while maintaining high-resolution imagery for crack detection purposes?

☐ Yes ☑ No

If yes, then what is the suitable range to capture effective images.

16. Should the system ensure the safety of inspection team?

☑ Yes ☐ No



Interview

Crack detection System in Concrete using AI drones

Interviewee: Isra Haider
Department: Software Engineering
Date: 30th November, 2023
Interviewed by: Maham Shahid
Interviewer Department: Software Engineering

About the system:

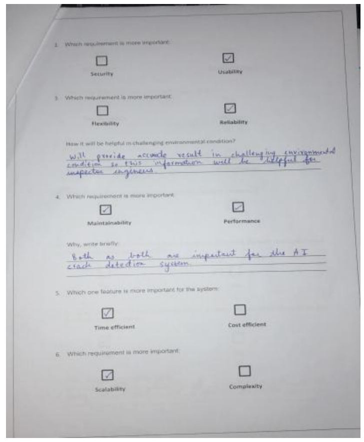
In this project, we are going to make a crack detection in concrete structure through AI for the purpose of infrastructure, such as buildings, bridges, pavement, etc needs to be examined periodically to maintain its reliability and structural health. Manual inspection is carried out by experienced inspectors who require long inspection times and rely on their empirical and subjective knowledge. To address this limitation, this study proposes a deep learning (DL)-based autonomous crack detection method using the convolutional neural network (CNN) technique. To improve the CNN classification performance for enhanced pixel segmentation, 40,000 RGB images were processed before training a pre-trained VGG16 architecture to create different CNN models. The Aim and objectives of our project are to provide an easier and cost-efficient approach to detect surface cracks in structure. Detect the crack. Analyze the crack and measure its depth and length. Use AI Drones to take images and analysing it, provide high resolution images and videos. Effective way to find sign of damages in structures without the danger of sending people to manually check sign of damages. We use control RGB, CNN, Python, MATLAB, Keras, Deep machine learning, and data set collection to make the system for the detection of cracks in concrete through AI Drones.

Give your opinion on the following questions and briefly describe your answers.

1. Which system will you prefer to use:

Manual crack detection system — Crack detection system using AI drones

As AI is the future of new development industries so this AI system will help the inspector engineers to detect cracks



2. Which requirement is more important:

Security — Usability

3. Which requirement is more important:

Flexibility — Reliability

How it will be helpful in challenging environmental condition?

will provide accurate result in challenging environmental condition so this information will be helpful for inspector engineers

4. Which requirement is more important:

Maintainability — Performance

Why, write briefly:

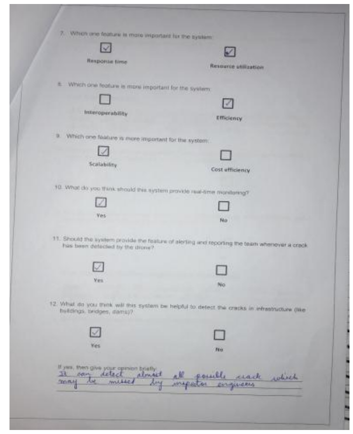
Both as both are important for the AI crack detection system

5. Which one feature is more important for the system:

Time efficient — Cost efficient

6. Which requirement is more important:

Scalability — Complexity



7. Which one feature is more important for the system:

Response time — Resource utilization

8. Which one feature is more important for the system:

Interoperability — Efficiency

9. Which one feature is more important for the system:

Scalability — Cost efficiency

10. What do you think should this system provide real-time monitoring?

Yes — No

11. Should the system provide the feature of alerting and reporting the team whenever a crack has been detected by the drone?

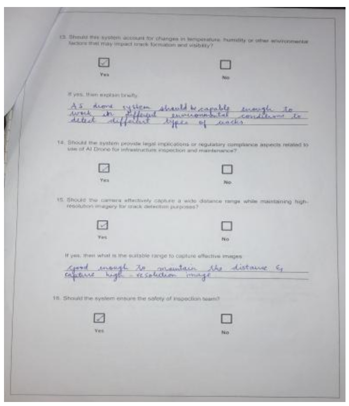
Yes — No

12. What do you think will this system be helpful to detect the cracks in infrastructure (like buildings, bridges, dams)?

Yes — No

If yes, then give your opinion briefly:

It can detect almost all possible crack which may be missed by inspector engineers



13. Should this system account for changes in temperature, humidity or other environmental factors that may impact crack formation and visibility?

Yes — No

If yes, then explain briefly:

AI drone system should be capable enough to work in different environmental conditions to detect different types of cracks

14. Should the system provide legal implications or regulatory compliance aspects related to use of AI Drone for infrastructure inspection and maintenance?

Yes — No

15. Should the camera effectively capture a wide distance range while maintaining high-resolution imagery for crack detection purposes?

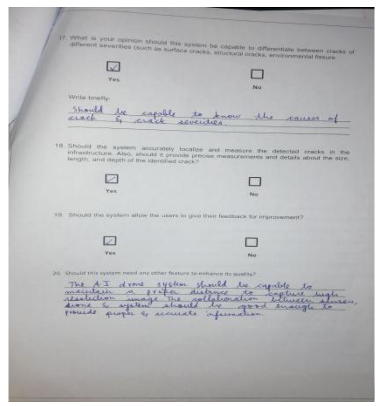
Yes — No

If yes, then what is the suitable range to capture effective images

Good enough to maintain the distance & capture high-resolution image

16. Should the system ensure the safety of inspection team?

Yes — No



17. What is your opinion should this system be capable to differentiate between cracks of different severities (such as surface cracks, structural cracks, environmental fissure

Yes — No

Write briefly:

Should be capable to know the causes of crack & crack severities.

18. Should the system accurately localize and measure the detected cracks in the infrastructure. Also, should it provide precise measurements and details about the size, length, and depth of the identified crack?

Yes — No

19. Should the system allow the users to give their feedback for improvement?

Yes — No

20. Should this system need any other feature to enhance its quality?

The AI drone system should be capable to maintain a proper distance to capture high resolution image. The collaboration between sensors, drone & system should be good enough to provide proper & accurate information

7. Functional Requirements:

The following are the functiona requirements of the system.

FR-01 The system shall detect cracks in infrastructures.

FR-02 The system should accurately localize the detected crack.

FR-03 The system shall provide precise measurements for identified cracks.

FR-04 The system shall accurately assess and classify the severity condition of detected cracks.

FR-05 The system shall be capable to know causes of cracks

FR-06 The system shall be able to differentiate between different types of cracks.

FR-07 The system shall provide Sensors for proper crack detection.

FR-08 The system shall capture a wide range distance.

FR-09 The system shall maintain high resolution.

FR-10 The system should account for changes in temperature.

FR-11 The system shall provide real time monitoring.

FR-12 The system shall provide the feature of alert and reporting the team.

FR-13 The system shall ensure each crack is registered and processed uniquely.

FR-14 The system shall store data for future use.

FR-15 The system should allow users to give feedback.

Non-Functional Requirement:

Non-functional requirements, also known as quality attributes or technical requirements. Non-functional requirements are critical aspects that describe the overall behavior and characteristics of a system rather than specific functionalities. In the context of crack detection in concrete through AI drones, some non-functional requirements might include:

1. **Accuracy:**
    - The system should achieve a high level of accuracy in detecting and classifying cracks in concrete structures.
    - The accuracy of the AI algorithm should be measurable and should meet or exceed predefined standards.

2. **Reliability:**
    - The crack detection system should be reliable, providing consistent results under different environmental conditions, such as varying lighting, weather, and surface textures.

3. **Scalability:**
    - The system should be able to scale to accommodate varying sizes of concrete structures, from small residential buildings to large infrastructure projects.

4. **Robustness:**
    - The AI algorithm and drone system should be robust enough to handle real-world challenges, such as different types of cracks, surface irregularities, and other anomalies that might be present in concrete structures.

5. **Performance:**
    - The system should operate within acceptable performance thresholds, providing timely and efficient crack detection without significant delays.
    - Considerations for the speed of data processing, drone movement, and communication between the drone and the central system are important.

6. **Autonomy:**
    - The drone system should exhibit a high degree of autonomy in terms of navigation, obstacle avoidance, and decision-making during the crack detection process.
    - It should be capable of adapting to dynamic environments without constant human intervention.

7. **Security:**
    - The system should have security measures in place to protect the collected data, especially if it involves sensitive information about infrastructure or buildings.

- Communication between the drone and the central system should be secure to prevent unauthorized access.
- The system shall provide security measures for the inspection team.

8. **Interoperability:**
   - The crack detection system should be compatible with different types of drones, sensors, and data storage systems.
   - It should be able to integrate seamlessly with existing infrastructure and workflows in construction and inspection processes.

9. **Usability:**
   - The user interface for interacting with the system should be intuitive, allowing operators to easily control and monitor the drone during crack detection missions.

10. **Maintainability:**
    - The system should be designed for ease of maintenance, with clear documentation, modular components, and the ability to update or replace components as needed.
    - Regular software updates and maintenance procedures should be straightforward and minimally disruptive.

8. Conclusion:

When developing a crack detection system for concrete using AI drones, considering these non-functional requirements is crucial to ensure the system's overall effectiveness and reliability in real-world applications.

9. Project Features:

i. **Detection Through Images:**

User can detect surface cracks in concrete structures through drones by uploading images and to webpage will show the highlighted image to the user on the webpage GUI.

ii. **Detection Through Videos:**

User also have the ability to detect surface cracks through drones by uploading a video to webpage GUI. The frames from the video will be passed individually from the model where it will highlight all the frames and convert the highlighted frames back into a video which will then be displayed to the user.

iii. **Download the Result:**

User will be able to download the image or video from the webpage as well.

iv. **Damages Type causes Crack:**

User can also detect the causes of surface cracks in concrete structures such as Earthquakes, Wind, Explosions, Temperature Variations, Moisture, permeability of concrete.

v. **Crack Detection of Different Types:**

User can observed different types of crack in concrete structures, such as Plastic-shrinkage cracking, Map cracking, Hairline cracking, Pop-outs, Scaling, Spalling,
D-cracking, Offset cracking, Diagonal corner cracking.

10. Software Requirement Specification (S.R.S)

i. Introduction:

The AI Drone Crack Detection System (ADCDS) represents an innovative and advanced system that combines state-of-the-art artificial intelligence (AI) algorithms with state-of-the-art drone technology to detect, analyze and monitor cracks in various infrastructure objects. By leveraging the capabilities of AI-powered algorithms and high-resolution drone sensors, this system aims to dramatically increase the accuracy, speed, and efficiency of crack detection methods. This Software Requirements Specification (SRS) document defines the functional and non-functional requirements needed to design, implement, and operate an AI drone crack detection system. It serves as a guiding framework in design, development, testing, and deployment stages, ensuring alignment with stakeholder requirements and industry standards.

Through this document, stakeholders will develop a comprehensive understanding of system quality, performance expectations, constraints, and compliance considerations.

Purpose:

DL methodologies and computer vision approaches are being extensively applied to automate crack discovery. One of the crucial approaches used for similar robotization is image processing (IP). Automated crack discovery using imagery reduces costs and assessment time and increases the safety and neutrality of concrete examinations. DL segmentation is the current trend in crack discovery, allowing the identification of pixels as cracks and thus allowing for crack size dimension.

Product Scope:

Following the objectives of the project, detecting surface cracks in concrete this service will be provided through a web based app. All the team members got together, we define the project's objectives, requirements, and deliverables. Some of the features and functionalities the software should include like crack detection, convolutional neural network, image processing, deep learning, damage detection, machine learning that we will build in this project.

ii. Overall Description:

Product Perspective: The ambitions of the project are

i) To develop and validate a CNN suitable for crack discovery.

ii) To train the developed CNN using recovered or unrefined images, creating different models.

iii) To analyze relative performance between the trained models in crack discovery.

Product Functions:

There are situations of crack discovery from images.

- The image is divided into patches and each patch is assigned a crack or non-crack marker
- A block is drawn around any detected crack
- Each pixel is labelled as crack or non-crack
- Dissect the crack and measure its depth and length
- Give high resolution images and vids.

Operating Environment: The operating atmosphere for crack discovery in concrete using AI drones can be complex and may involve the following factors:

Drones (UAVs): Unmanned Aerial Vehicles (UAVs) equipped with cameras and detectors are essential for attaining high- resolution images and data of concrete structures.

Camera Systems: High- resolution cameras, including visible light and infrared cameras, are generally used for attaining images of concrete exteriors.

AI Algorithms: Artificial intelligence algorithms, particularly deep learning models, are employed for image analysis and crack discovery.

Weather Considerations: The operating climate includes rainfall conditions similar as wind, rain, and temperature.

User Interface: An intuitive interface for operators to control the drone, cover data in real-time, and interpret results is vital.

Maintenance and Calibration: Regular conservation of drones and estimation of detectors is necessary to insure accurate and dependable data collection over time.

Design and Implementation Constraints:

Advantages:

- It is not time consuming
- Un-expensive
- Easy to install
- Better Quality of the picture of crack

Limitation:

- Only one image or video can be uploaded at a time
- Detect single crack at a time

iii. External Interface Requirements

User Interfaces (UI): User Interfaces include

Live Video Feed: Display a live videotape feed from the drone's cameras, allowing drivers to see the concrete structure in real- time.

Flight Controls: Include intuitive controls for managing the drone's flight. This may involve buttons or a joystick for automatic control, as well as automated flight path options for predefined inspection routes.

Navigation Map: give a chart interface that shows the drone's current position and its flight path. This helps drivers imagine the examination area and ensures the drone is covering the entire structure.

Crack Analysis Dashboard: This dashboard should display details about detected cracks, similar as size, position, and severity. Color coding or heat-maps can be used to emphasize areas with possible issues.

Alerts and announcements: apply a system for real- time cautions and notifications. However, the driver should be in-continently notified through visual and auditory cues, if the AI algorithm detects a significant crack or structural issue.

Data Export: Allow drivers to export examination reports, including images, analysis results, and any other applicable data. This is essential for documentation, analysis, and collaboration with other stakeholders.

Intuitive Design: Keep the user interface simple, intuitive, and user-friendly.

Hardware Interface:

Detecting cracks in concrete using AI-powered drones involves the integration of various hardware interfaces to capture, process, and analyze data.

1. Drone Platform:
    - The drone serves as the aerial platform for data collection. Multi-rotor drones are commonly used for their stability and maneuverability, while fixed-wing drones may cover larger areas.
2. Sensors:
    - RGB Cameras: High-resolution RGB cameras capture visual information of the concrete surface. This data is useful for creating visual maps and for contextual analysis.
3. Communication Systems:
    - Wireless Communication Module: Enables real-time communication between the drone and the ground station for live monitoring and control.
    - Data Transmission Systems: High-bandwidth communication systems are essential for transmitting large amounts of data collected by sensors back to the ground station.

4. Embedded AI Processor:

- A dedicated AI processing unit onboard the drone allows for real-time analysis of collected data. This can include GPUs or specialized AI chips for efficient processing of machine learning algorithms.

Software Interfaces:

Drone Control Software Interface:

API (Application Programming Interface): Define an API that allows the control software to transfer commands to the drone for navigation, flight path planning, and obtaining images.

API for AI Models: Develop an API that enables communication between the AI algorithm and other software factors. This includes transferring input images for analysis and taking the results.

Communication Protocol: Specify the communication protocol used between the drone and the ground station software for real- time data transmission.

API for Data Retrieval: Develop an API for the ground station software to recoup data from the drone, including images, analysis results, and other applicable information. Database Interface: Database API Define an interface for storing and reacquiring data related to inspections, including historical examination records, images, and analysis results.

Communication Interface:

1. Data Encryption and Security Protocols:
    - To ensure the security of transmitted data, communication interfaces should implement encryption protocols. This is critical when dealing with sensitive infrastructure information.
2. Remote Control Systems:
    - Manual control of the drone is often necessary, and a remote controller provides a physical interface for a human operator to control the drone manually.
3. Video Transmitters:

- For applications requiring live video feeds, drones may use video transmitters to stream real-time footage to the ground control station. This aids in visual inspection and monitoring.

11. System Features

Damages Causes Types:

Infrastructure situation getting worse and worse whether bridges, dams, buildings etc. They need proper intervention to guarantee their durability and services life. The following are the possible causes of cracks in concrete structures, such as

- Earthquakes
- Wind
- Explosions
- Temperature Variations
- Moisture
- Permeability of concrete

This is defined by an ensemble of cracks, scaling, spalling, and efflorescence.

Crack Detection:

The cracking splendor in concrete is complex, and it depends on a number of factors similar as rate of drying, measure of drying, tensile strength and strain, flexibility, drying loss, and degree of restraint. Concrete contains further water than demanded for hydration thus loss begins when concrete starts to dry. No cracks are developed when concrete is unrestrained, but it isn't possible to support structures without conditions. Different types of crack are observed in concrete structures similar as

- Plastic- shrinkage cracking,
- Map cracking,
- Hairline cracking,
- Pop- outs
- Scaling
- Spalling

- D- cracking
- Offset cracking
- Diagonal corner cracking.

The structural stability and continuity aren't affected by utmost types of cracking still, in extreme cases, it may be affected. Set up images converted to grayscale increased the performance of a CNN with object recognition.

Methodologies:

Dataset Collection:

Focal distance of 50 cm use to take picture of cracks. The images were taken vertical to the face. The training dataset was used to train the model and the confirmation dataset was used during training to cover the model learning curve. The datasets contains images of colorful concrete surfaces with and without crack. The image data are divided into two as negative (without crack) and positive (with crack) in separate brochure for image category. Each class has 20000 images with a aggregate of 40000 images with 227 x 227 pixels with RGB channels. The dataset is generated from 458 high- resolution images (4032x3024 pixel). High resolution images set up out to have high variance in terms of exterior finish and illumination condition. No data addition in terms of arbitrary rotation or flipping or tilting is applied.

Image Processing:

Prepare the images for analysis by applying preprocessing techniques, such as resizing, normalization, and color adjustments.

i. The images were processed separately once imported using Tensor-Flow and were converted from images to a sequence of elements. The images were processed to obtain four datasets: RGB, grayscale, edge detection, and binarization.

Control (RGB):

Standard colour images were used in CNN image categorization. These images featured values for each of the primary colours on the range red, green, and blue, also known as 3- channels (control

RGB). A combination of these three can produce all possible colour pallets. The image is made up of multiple pixels, with every pixel conforming of three different values for the RGB channels. The values of these channels and the channels of the embracing pixels were used to identify cracks.

Deep Learning-Convolutional Neural Network (CNN): The most common method of DL for crack detection is CNN. CNNs have been shown to outperform edge detection and ML classifiers. Pre-trained CNNs have previously demonstrated an accuracy of over 98% for crack/non-crack detection. However, the CNN can only identify the image with the crack, not the crack as a pixel or feature. CNNs are particularly effective for image-based tasks like crack detection.

Model Development:

The code was programmed using Python with Keras and TensorFlow python packages.

1) Main Architecture of the CNN is use library Tensorflow+Keras which were proven to be very efficient in dealing with images and coding of algorithms was performed by python.

2) KERAS:

The Keras package contains numerous DL infrastructures. The Keras VGG16 model is grounded on the model proposed. A non-crack was labelled 0, while a crack was labelled 1. The program rounded down to predict a crack, a function was created to convert figures ≥0.5 to 1. This allowed the program to determine if an image was a crack or not. Standard colour images were used in CNN image categorization. These images featured values for each of the primary colours on the diapason red, green, and blue. A combination of these three can produce all possible colour pallets. The image is made up of multiple pixels, with every pixel conforming of three different values for the RGB channels. The values of these channels and the channels of the embracing pixels were used to identify cracks.

3) MATLAB:

- The crack detection algorithm was proven to deliver better appearance of thin hairline cracks and shear cracks by using MATLAB approach.

- Isolated white pixels or noise in the cracks images was able to be removed when the image undergo all the filtration process.

MATLAB Removes:

- Background Noise
- Unwanted information
- Illumination

Feature Extraction:
Allow the model to learn features that distinguish between cracked and non-cracked regions. This step is typically handled automatically by the deep learning model during the training process.

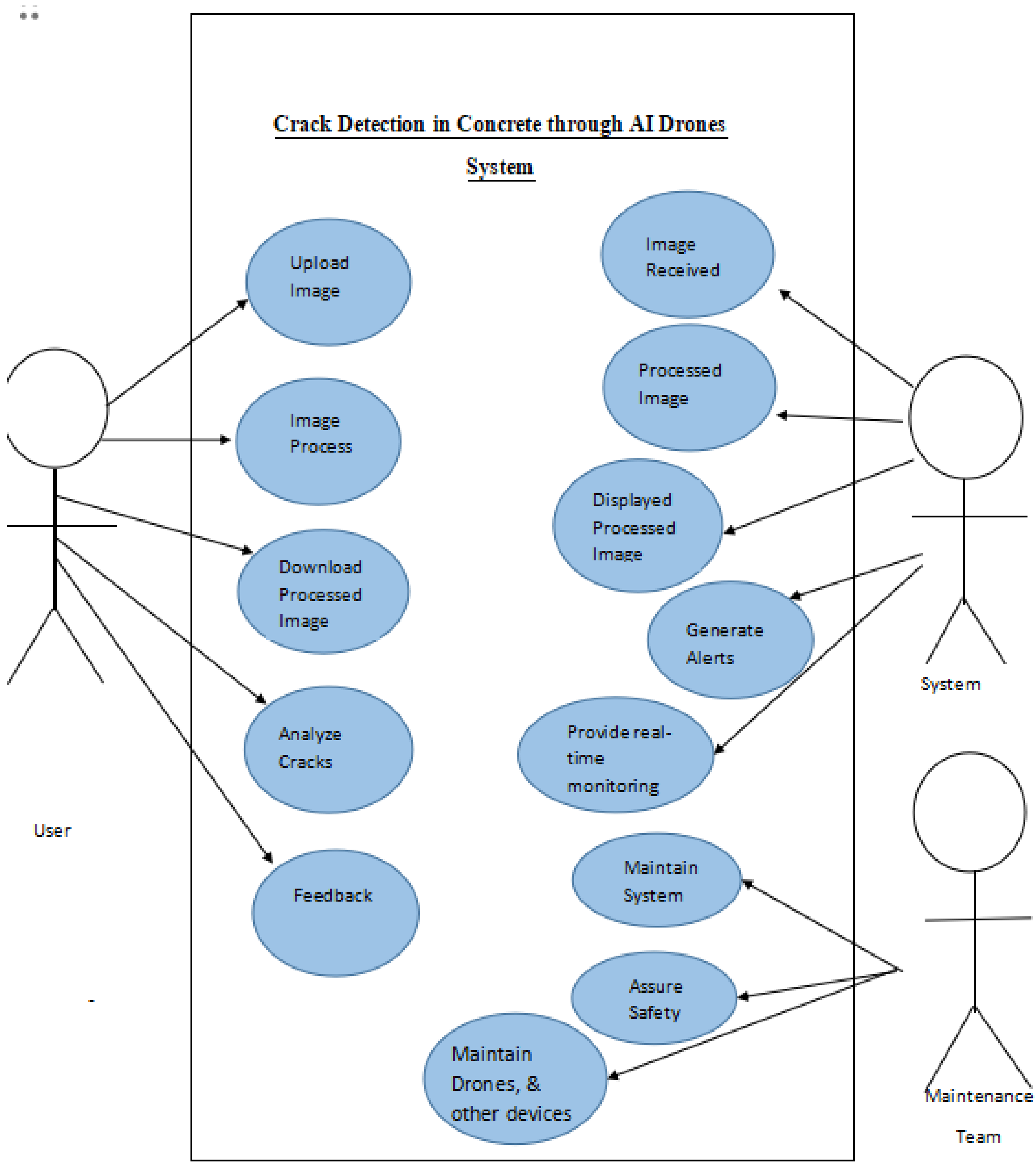
Deployment:
Deploy the trained model for real-time crack detection. This could involve integrating the AI system with cameras or sensors positioned to monitor the concrete structure.
Continuous Monitoring:
Implement a continuous monitoring system where the AI model regularly analyzes new data to detect any emerging cracks or changes in the structure.

Alerts and Reporting:
Set up a system to generate alerts or reports when the AI model identifies cracks beyond a certain threshold. This enables timely intervention and maintenance.

Feedback Loop:
Establish a feedback loop where the system continually improves by incorporating new data and updating the model periodically.

It's important to note that the success of an AI-based crack detection system depends on the quality and diversity of the training data, the chosen algorithm, and the system's ability to adapt to different environmental conditions. Additionally, collaboration with domain experts, such as structural engineers, is essential to ensure the accuracy and reliability of the AI system.

12. Use Case Diagram:



1) Upload Image:

UC-01	Upload Image	
Actor	User	
Goal	User should be able to upload image in the system.	
Pre-Request	User is authenticated and has access to the system. The system is operational and capable of receiving image files	
Description	<u>User Action:</u> User selects option to upload an image. User chooses the desired image file	<u>System Response:</u> Accept or reject based on file format. Receive and store the uploaded image
Alternate	If the selected file is not in an accepted format (e.g., JPEG, PNG), the system displays an error message and prompts the user to select a valid image file.	

2) Images Processing:

UC-02	Image Processing
Actor	User
Goal	Analyze the uploaded images to identify and locate cracks.
Pre-Request	Images are successfully uploaded and stored in the system. AI algorithm for crack detection is operational.

Description	<u>User Action:</u> N/A (System performs the processing automatically after image upload)	<u>System Response:</u> System analyze the image. Apply AI algorithms to scan and detect cracks within the image.
Alternate	If the uploaded image quality is too low, the system notifies the user of poor quality and requests a higher quality image	

3) Download the Processed Image:

UC-03	Processed Image	
Actor	User	
Goal	Obtain the processed image with identified cracks for further analysis or documentation.	
Pre-Request	The image has been processed by the AI system, and the analysis results are available. User has appropriate permissions to download the processed image.	
Description	<u>User Action:</u> Selects the option to download the processed image. Clicks on the download link provided by the system	<u>System Response:</u> Provide a download link or option for the processed image with marked crack locations.
Alternate	If the system encounters an error during download (e.g., file corruption), it notifies the user and prompts for re-download. If any unauthorized thing happen then don't login the system.	

4) Crack Analysis:

UC-04	Crack Analysis

Actor	User	
Goal	Conduct an in-depth analysis of identified cracks to assess severity, dimensions, and potential impact.	
Pre-Request	The processed image with identified cracks is available for analysis.	
Description	User Action: User must analyze the crack images received by the system.	System Response: Utilize specialized tools or algorithms to examine identified cracks in the processed image.
Alternate	If the system encounters errors while analyzing cracks (e.g., insufficient data for analysis), it notifies the user and prompts re-examination or provides guidance. If any unauthorized thing happen then don't login the system.	

5) Feedback:

UC-05	Feedback	
Actor	User	
Goal	Provide feedback to the AI system regarding the accuracy or usefulness of crack detection and analysis results.	
Pre-Request	The user has interacted with the crack detection system, and the analysis results are available for review.	
Description	User Action: User must give the feedback about the system.	System Response: Display an interface or prompt asking the user to provide feedback on the crack detection results. System stores the feedback.
Alternate	If the user intends to provide detailed feedback but lacks a platform to do so, the system may prompt or guide the user to the appropriate feedback submission channel.	

6) Image Received

UC-06	Image Received	
Actor	System	
Goal	Receive images from users for crack detection analysis.	
Pre-Request	The system is operational and ready to receive image files.	
Description	System Action: Accept image files uploaded by users for crack detection analysis.	User Response: Accept image files uploaded by users for crack detection analysis.
Alternate	If the system encounters issues in receiving images (e.g., network error), it notifies the user of the problem and provides guidance or troubleshooting steps.	

7) Image Processed:

UC-07	Image Processed
Actor	System
Goal	Obtain processed images after crack detection for analysis or documentation.
Pre-Request	The processed image with identified cracks is available and accessible within the system.

Description	System Action: Displays or provides access to the processed image post-crack detection analysis	User Response: Requests the system to retrieve and display the processed image after the crack detection analysis.
Alternate	If the system encounters errors in displaying the processed image (e.g., image rendering issues), it notifies the user and provides troubleshooting steps.	

8) Displayed Processed Image:

UC-08	Displayed Processed Image	
Actor	System	
Goal	View the processed image with identified cracks for analysis or documentation purposes.	
Pre-Request	The processed image with identified cracks is available and accessible within the system.	
Description	System Action: System processed the image and display the processed image with marked cracks..	User Response: Accesses the system interface or designated platform to view the processed image.
Alternate	If the system encounters errors in displaying the processed image	

9) Generate Alerts

UC-09	Generate Alerts
Actor	System

Goal	Alert users or system administrators about critical or severe crack detection findings that require immediate attention.	
Pre-Request	The system has successfully detected critical cracks or anomalies that warrant an alert.	
Description	System Action: System received image then it generate alerts to designated users or administrators, providing information about the detected critical cracks.	User Response: User respond towards the alerts.
Alternate	If any unauthorized thing happen then don’t use the system.	

10) Provide Real-Time Monitoring:

UC-10	Provide Real-Time Monitoring	
Actor	System	
Goal	Monitor crack detection and analysis processes in real-time for immediate insights and actions.	
Pre-Request	The system is operational, and crack detection processes are actively running.	
Description	System Action: System provide real-time monitoring.	User Response: User analyze the real-time monitoring observe ongoing crack detection activities and take immediate actions if required.
Alternate	If any unauthorized thing happen then don’t use the system.	

11) Maintain System:

UC-11	Maintain System	
Actor	Maintenance Team	
Goal	Ensure the smooth operation and upkeep of the AI Crack Detection System by performing maintenance tasks and addressing system issues.	
Pre-Request	Access permissions granted to the maintenance team, knowledge of system architecture, and established maintenance procedures.	
Description	Maintenance Team Action: Conduct routine checks on hardware and software components.	System Response: Allow access to system components and tools necessary for maintenance tasks, such as software updates, hardware checks, or database optimizations.
Alternate	If the maintenance process encounters unexpected issues or complications, the team follows contingency plans or contacts specialized support for resolution. If any unauthorized thing happen then maintenance team take action.	

12) Assure Safety:

UC-12	Assure Safety
Actor	Maintenance Team
Goal	Ensure the safety and reliability of the AI Crack Detection System for operation and maintenance
Pre-Request	Maintenance team authorization for maintenance tasks, established safety protocols, and access to safety documentation.

Description	Maintenance Team Action: Assure the Safety of the System.	System Response: Provide safety guidelines, protocols, and training materials for users and maintenance teams.
Alternate	If safety protocols are breached or if unsafe conditions are identified, both the user and maintenance team take immediate action to rectify the situation and ensure safety.	

13) Maintain Drones & System:

UC-13	Maintain Drones & System	
Actor	Maintenance Team	
Goal	Ensure the operational functionality both the drones and the AI Crack Detection System used in conjunction with them.	
Pre-Request	User and maintenance team authorization, access to maintenance tools/resources, established maintenance procedures for drones and system components.	
Description	Maintenance Team Action: Maintain the functioning of drones and system. If issues arise, take corrective action, and advise the user on necessary operational adjustments.	System Response: Support the maintenance team with necessary data and resources for efficient maintenance.
Alternate	If maintenance reveals issues with drones or system components, the maintenance team takes immediate action to address the problem and, if necessary, advises the user on temporary operational adjustments.	

13. Requirement Validation Checklist:

SR No:	Requirement Statement	Yes	No
FR-01	The system shall detect cracks in infrastructures.	✓	
FR-02	The system should accurately localize the detected crack.	✓	
FR-03	The system shall provide precise measurements for identified cracks.	✓	
FR-04	The system shall accurately assess and classify the severity condition of detected cracks.	✓	
FR-05	The system shall be capable to know causes of cracks	✓	
FR-06	The system shall be able to differentiate between different types of cracks.	✓	
FR-07	The system shall provide Sensors for proper crack detection.	✓	
FR-08	The system shall capture a wide range distance.	✓	
FR-09	The system shall maintain high resolution.	✓	
FR-10	The system should account for changes in temperature.	✓	
FR-11	The system shall provide real time monitoring.	✓	
FR-12	The system shall provide the feature of alert and reporting the team.	✓	

FR-13	The system shall ensure each crack is registered and processed uniquely.	✓	
FR-14	The system shall store data for future use.	✓	
FR-15	The system should allow users to give feedback.	✓	
NFR-01	The system shall maintain operational reliability in various weather conditions (like low breeze, cloudy, sunny, drizzle)	✓	
NFR-02	The system shall provide security measures for the inspection team.	✓	
NFRS-03	The system shall offer maintainability, ensuring easy servicing and updates.	✓	
NFR-04	The system shall effectively utilize available resources.	✓	
NFR-05	The system shall comply with legal implications and regulatory standards.	✓	
NFR-06	The system should demonstrate efficiency in its operations.	✓	
NFR-07	The system shall be time efficient.	✓	
NFR-08	The system shall provide scalability features.	✓	

10) Requirement Traceability Matrix:

Requi ID	Requirement Description	System Architecture	Hardware	Software	Test Plans	Drone Firm-ware	AI Model Training Data	Crack Detection Algorithm	Planning Software	Data Storage Specification	Interface Design	Assurance Plan	Verification
R1	Detect Crack	X	X	X	X	X	X	X	X	X		X	X
R2	Accurately Localize crack	X	X	X	X	X	X	X	X	X		X	X
R3	Provide Precise Measurement	X	X	X	X	X	X	X	X			X	X
R4	Classify the Severity Condition	X		X	X		X	X	X		X	X	X
R5	Capable to know causes of Cracks	X		X	X		X	X	X		X	X	X
R6	Differentiate between different Cracks	X		X	X		X	X	X		X	X	X
R7	Sensors for Proper Detection			X	X	X	X	X				X	X
R8	Capture Wide Range Distance			X	X	X	X	X				X	X
R9	Maintain High Resolution	X		X	X	X	X	X	X		X	X	X
R10	Accounts for changes Temperature		X	X	X	X	X	X				X	X
R11	Real time Monitoring	X		X	X	X	X	X			X	X	X
R12	Alert Generating	X		X	X	X	X	X	X		X	X	X
R13	Ensure Crack is Registered	X		X	X	X	X	X	X	X	X	X	X
R14	Data Storage	X		X	X	X	X	X	X	X	X	X	X
R15	Feedback	X	X	X	X	X	X	X			X	X	X

- "X" indicates a link between the requirement and the corresponding artifact.

- The columns represent different phases or components of the project, including System Architecture, Hardware, Software, Test Plan, Drone Firmware, AI Model Training Data, Crack Detection Algorithm, Data Storage Specification, User Interface Design, and Quality Assurance Plan.

- This traceability matrix helps ensure that each requirement is addressed by linking it to the specific artifacts or components responsible for its implementation. It provides transparency and facilitates project management by tracking the progress of each requirement throughout the development lifecycle.

Dependency of Functional Requirements on one another:

INDEPENDENT	DEPENDENT						
	FR-02	FR-03	FR-04	FR-05	FR-06	FR-09	FR-13
FR-01	×	×	×	×	×		×
FR-07							
FR-08						×	
FR-11							
FR-11							
FR-12							
FR-14							×
FR-15							

The results of above requirement traceability matrix are shown as intersection of the FRs. A **“X”** is used to show the intersection. Functional Requirement (FR)-01 is associated with FR-02, FR-03, FR-04, FR-05, FR-06 and FR-13, demonstrating that these requirements have dependency on the FR-01. FR-07, however, doesn’t directly relate to any other requirement, standing as an independent specification. FR-08 is linked to FR-09, showcasing a relationship where FR-09 relies on FR-08 for its fulfillment. The remaining requirements, FR-11, FR-12, and FR-15 don't exhibit direct dependencies on other specified functional requirements. FR-13 is linked to FR-01 and FR-04, indicating that FR-13 is reliant on both FR-01 and FR-04 for its fulfillment. This matrix highlights the interdependencies and connections between various functional requirements, illustrating their relationships and dependencies on each other for successful implementation.